

A Cortico-Cerebellar Model for Solving Non-Markovian Environments



Universidad del Desarrollo

Luis Duarte¹, *Principal Investigator*² (ORCID: 0000-0000-0000-0000)

¹Centro de Investigación en Complejidad Social (CICS), Universidad del Desarrollo, Santiago, Chile

²Department, Center, City, Country



CENTRO DE INVESTIGACIÓN EN COMPLEJIDAD SOCIAL

INTRODUCTION

⊢ Cerebellar lesions spare delay conditioning but abolish trace conditioning, revealing a failure to bridge temporal gaps.

⊢ Navigating these non-Markovian gaps requires delayed credit assignment, yet canonical models treat the cerebellum as purely feedforward.

⊢ Feedforward architectures trap the cortex in a Markovian regime restricted to instantaneous observations.

Hypothesis. Passing cortical states through a granular expansion, followed by sparse compression and thalamic return, provides the necessary temporal integration to resolve non-Markovian delays without backpropagation-through-time.

RESULTS

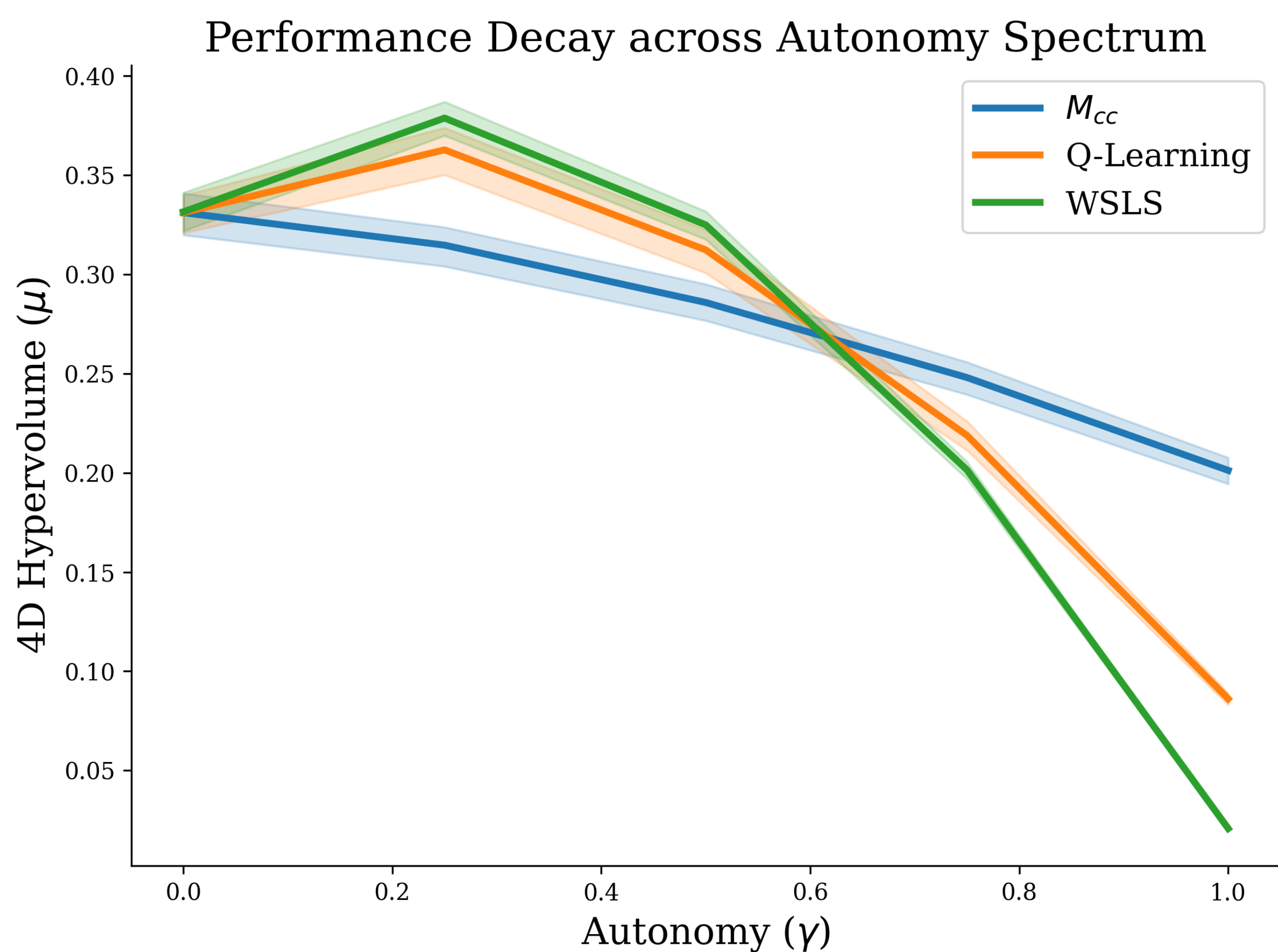


Figure: **NSGA-II Discovery.** Multi-objective evolution jointly minimized distillation and empirical loss to converge on structural bounds.

RESULTS (CONT.)

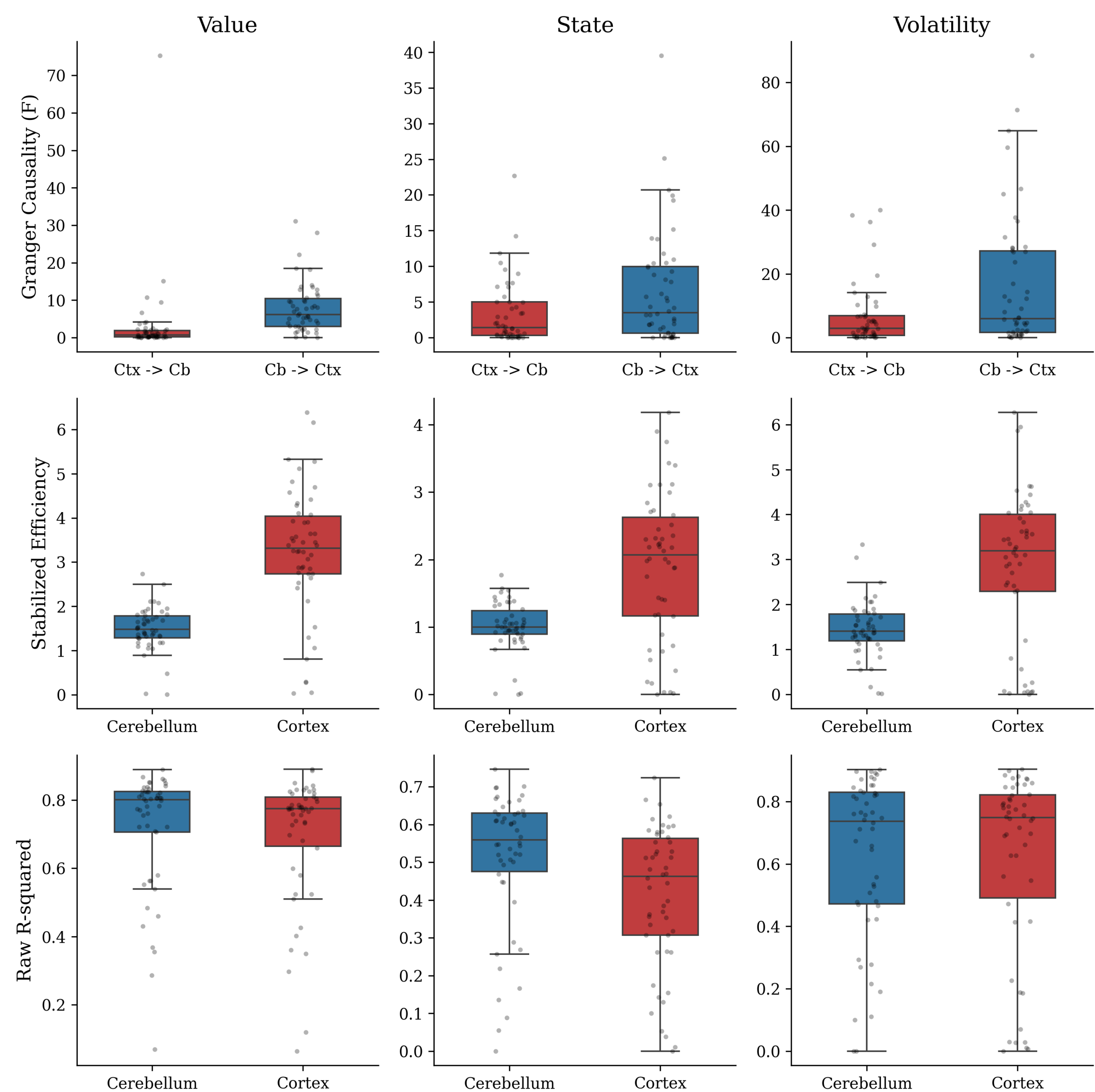


Figure: **Directional Flow.** Granger causality reveals significant causal flow from Cerebellum to Cortex across Value, State, and Volatility.

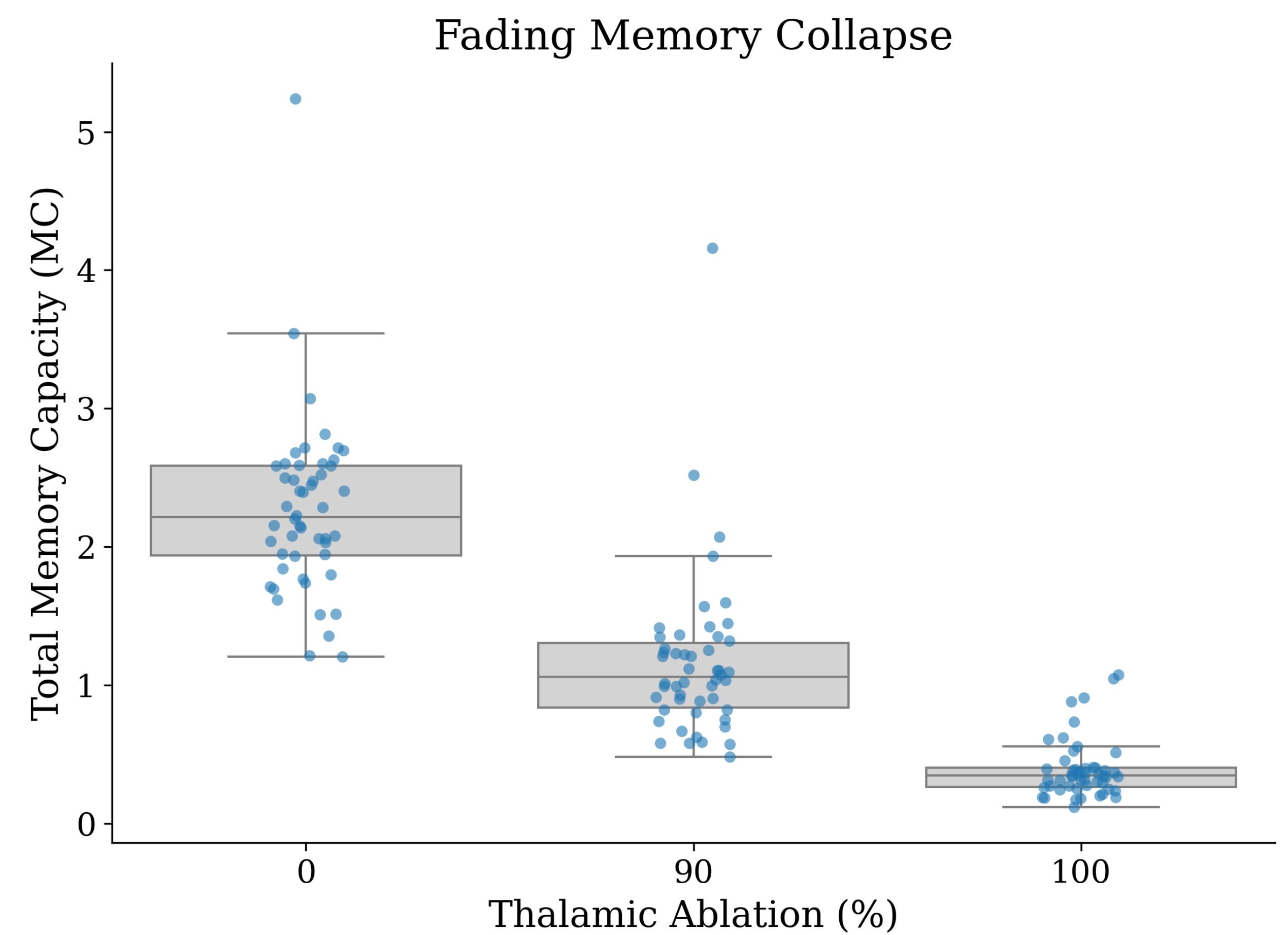


Figure: **Temporal Integration.** The baseline M_{cc} maintains temporal memory, while complete thalamic ablation collapses into a Markovian reactor.

METHODS

State Expansion & Fading Memory. The granular layer acts as an Echo State Network to generate an eligibility trace.

$$\phi(X_t) = W_{\text{exp}} X_t \quad (1)$$

Where X_t is the cortical state, W_{exp} is the fixed immersion matrix, and $\phi(X_t)$ is the expanded trace.

Sparse Compression. The DCN compresses the trace without geometric collapse.

$$C_t = P_{\theta}(\phi(X_t)) \quad (2)$$

Where P_{θ} is a sparse Achlioptas projection preventing